



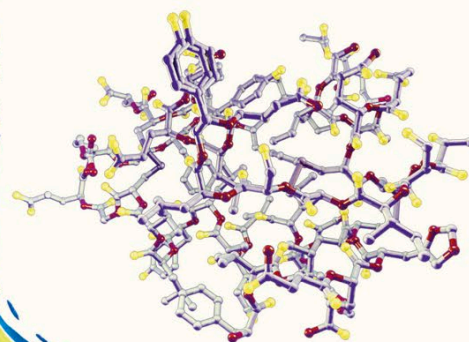
Dorothy suffered from a disease called rheumatoid arthritis for most of her life. Despite worsening joint pain, she made models and X-ray images of key chemical structures. Some of these images showed very complex patterns for vitamin B12 crystals. Dorothy used one of the very first computers to help her identify the crystal's 1,000 atoms.

Mapping molecules

X-ray crystallography involves shooting X-rays through crystals of a pure chemical. The patterns formed by the X-rays are like a map of the chemical. Scientists who learn how to read these maps can figure out which atoms its molecules are made from, and how they fit together. Dorothy became an expert at deciphering the most mind-bogglingly complex chemicals of all—those made by living things. Once scientists fully understand a molecule, they can figure out how it works, which can lead to new treatments and medicines. Dorothy also worked to improve science education and encouraged other women to follow her into the colorful world of chemistry.

Insulin

After 30 years, Dorothy had improved X-ray crystallography so much that she was able to crack her toughest problem yet—the structure of insulin. This is a substance made by the body. Without enough of it, people suffer from a disease called diabetes.



Dorothy won a Nobel Prize in Chemistry for her work on vitamin B12. She was also the second woman in history to receive the British Order of Merit.